DNS AND DHCP TROUBLESHOOTING ASSIGNMENT

1. Simulate a DNS resolution failure by editing the hosts file to point www.spotify.com to an incorrect IP address. Ping the domain and document the error message.

Procedure

1. Open the hosts file with administrator privileges. On Windows, its location is C:\Windows\System32\drivers\etc\hosts; on Linux, it is /etc/hosts.
2. Add a temporary entry using the documentation-only test address shown below.
3. Save the file, then run the ping command in a terminal.
4. After recording the result, remove the temporary entry and save the hosts file to restore normal name resolution.

Hosts entry: 203.0.113.25 www.spotify.com

Command to run: ping www.spotify.com

ping www.spotify.com
Pinging www.spotify.com [203.0.113.25] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.25:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Important: 203.0.113.25 is reserved for documentation and should not be used as a real public server address. Remove the hosts-file entry immediately after the test.

2. Configure a DHCP server in a network simulator and intentionally set an incorrect default gateway. Connect a client, verify the assigned gateway, and explain how to identify and fix the issue.

Setting	Values
Network	192.168.20.0/24
DHCP server	192.168.20.2
DHCP address pool	192.168.20.100 – 192.168.20.150
Correct gateway	192.168.20.1
Incorrect gateway configured in DHCP	192.168.20.254
Example client address	192.168.20.100

How to identify the issue: Check the client's IP configuration using ipconfig (Windows) or ip route (Linux). The client will show 192.168.20.254 as its default gateway, which does not match the router's actual address of 192.168.20.1. The client may communicate within its local subnet but will usually fail to reach other networks.

How to fix it: Open the DHCP pool settings and change the default-router / gateway option to 192.168.20.1. Renew the client's DHCP lease (for example, use ipconfig /release followed by ipconfig /renew on Windows) and verify that the correct gateway is now assigned.

3. Devices can access websites by IP address but not by domain name. Use nslookup or dig to diagnose the DNS problem and document the troubleshooting steps and solution.

Commands

```
nslookup www.google.com
nslookup www.google.com 8.8.8.8
```

Troubleshooting steps: 1. Run nslookup to check whether the configured DNS server can resolve a domain. 2. Note the DNS server address and any timeout or 'server can't find' error. 3. Query a known public DNS resolver such as 8.8.8.8 for comparison. 4. If the public resolver works but the configured resolver fails, check the device's DNS settings and the local DNS server. 5. Correct the DNS server address or repair the DNS service, then repeat the lookup.

nslookup www.google.com

Server: Unknown Address: 192.168.20.10

DNS request timed out.

timeout was 2 seconds.

DNS request timed out.

timeout was 2 seconds.

nslookup www.google.com 8.8.8.8

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: www.google.com

Addresses: 142.250.183.100

2404:6800:4009:82a::2004

4. A device is not receiving an IP address from DHCP and shows an APIPA address (169.x.x.x). List possible causes and explain how to resolve the issue.

Possible causes

- The DHCP server is switched off, stopped, or unreachable.
- The DHCP address pool is exhausted and no free IP addresses remain.
- A network cable, Wi-Fi connection, switch port, or VLAN configuration is faulty.
- A firewall or network setting is blocking DHCP traffic.
- The DHCP scope or subnet settings are configured incorrectly.

Step-by-step resolution

5. Check the physical or Wi-Fi connection and confirm that the client is connected to the correct network.
6. Check whether the DHCP server is running and reachable from the client.
7. Inspect the DHCP scope and confirm that free addresses are available.
8. Verify that the DHCP scope, subnet mask, VLAN, and relay configuration (if used) are correct.
9. Check firewall rules and ensure DHCP traffic is not blocked.

10. Renew the client's lease using ipconfig /release and ipconfig /renew on Windows, or reconnect to the network; then verify that it receives a valid address.

Explanation: An APIPA address in the 169.254.0.0/16 range is self-assigned when a device cannot obtain an IPv4 address from a DHCP server. It usually indicates that DHCP communication or address allocation has failed.

5. Create a table documenting at least three common DNS issues and three common DHCP issues, including symptoms, causes, and recommended solutions.

Type	Issue / symptom	Possible cause	Recommended solution
DNS	Website opens by IP address but not by name.	Incorrect or unreachable DNS server.	Configure a working DNS server and test again with nslookup or dig.
DNS	DNS lookup times out.	DNS service is down, blocked, or unreachable.	Check server status, network connectivity, and firewall rules for port 53.
DNS	A domain resolves to an incorrect IP address.	Wrong or outdated DNS record or local hosts-file entry.	Correct the record, remove the incorrect hosts entry, and clear the DNS cache if needed.
DHCP	Client receives a 169.254.x.x APIPA address.	DHCP server is unreachable or DHCP requests are blocked.	Check connectivity, DHCP service, VLAN/relay, and firewall settings; renew the lease.
DHCP	Client cannot obtain an address; pool has no free leases.	DHCP scope is exhausted.	Expand the address pool or release unused leases.
DHCP	Client receives an incorrect default gateway.	Wrong gateway option configured in the DHCP scope.	Correct the DHCP scope's router option and renew the client lease.

Conclusion: DNS translates domain names into IP addresses, while DHCP automatically provides IP configuration to clients. Checking command output, server settings, address pools, and network connectivity helps identify and resolve common DNS and DHCP problems.